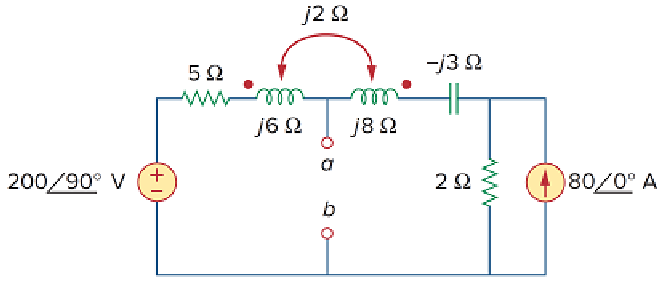
Chapter 13, Problem 14P

Problem

Obtain the Thevenin equivalent circuit for the circuit in Fig. 13.83 at terminals *a*-*b*.

Figure 13.83



Step-by-step solution

Step 1 of 8

Refer to the circuit diagram in Figure 13.83 in the textbook.

Convert the current source $80\angle 0°\ \text{A}$ into voltage source using source transformation.

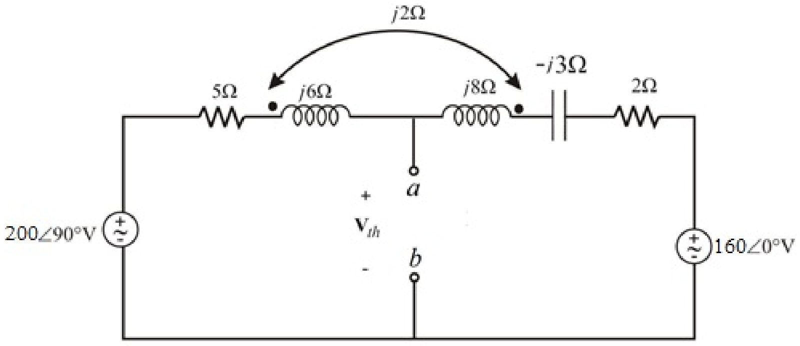


Figure 1

Step 2 of 8

Apply Kirchhoff's voltage law around outer loop.

$$-200\angle 90° + 5\mathbf{I} + (j6 - j8 - j4)\mathbf{I} + (-j3 + 2)\mathbf{I} + 160\angle 0° = 0$$
$$(7 + j7)\mathbf{I} + 160 - j200 = 0$$

Solve for $\mathbf{I}$.

$$\begin{aligned}\mathbf{I} &= -\frac{160 - j200}{7 + j7} \\ &= \frac{256.125\angle 128.66°}{9.9\angle 45°} \\ &= 25.87\angle 83.66° \text{ A}\end{aligned}$$

Comments (2)

Anonymous

i think there might be an arithmetic error in this step. i got (7-j9)I+160-j200=0

Anonymous

/\ nevermind it should be +j8 in the first line

Step 3 of 8

Apply Kirchhoff's voltage law around left loop.

$$-200\angle 90° + 5\mathbf{I} - j6\mathbf{I} - j2\mathbf{I} + \mathbf{V_{Th}} = 0$$
$$-200\angle 90° + \mathbf{I}(5 + j4) + \mathbf{V_{Th}} = 0$$

Substitute $25.87\angle 83.66° \text{ A}$ for $\mathbf{I}$.

$$-200\angle 90° + (25.87\angle 83.66° \text{ A})(6.40\angle 38.65°) + \mathbf{V_{Th}} = 0$$
$$-200\angle 90° + 165.66\angle 122.31° + \mathbf{V_{Th}} = 0$$
$$-j200 - 88.57 + j140 + \mathbf{V_{Th}} = 0$$
$$\begin{aligned}\mathbf{V_{Th}} &= 88.57 + j60 \text{ V} \\ &= 107\angle 34.11° \text{ V}\end{aligned}$$

Thus, the Thevenin's voltage $\mathbf{V_{Th}}$ is $\boxed{107\angle 34.11° \text{ V}}$.

Step 4 of 8

To obtain $\mathbf{Z}_{th}$ insert 1A current source as test source and replace all the voltage sources with short circuit.

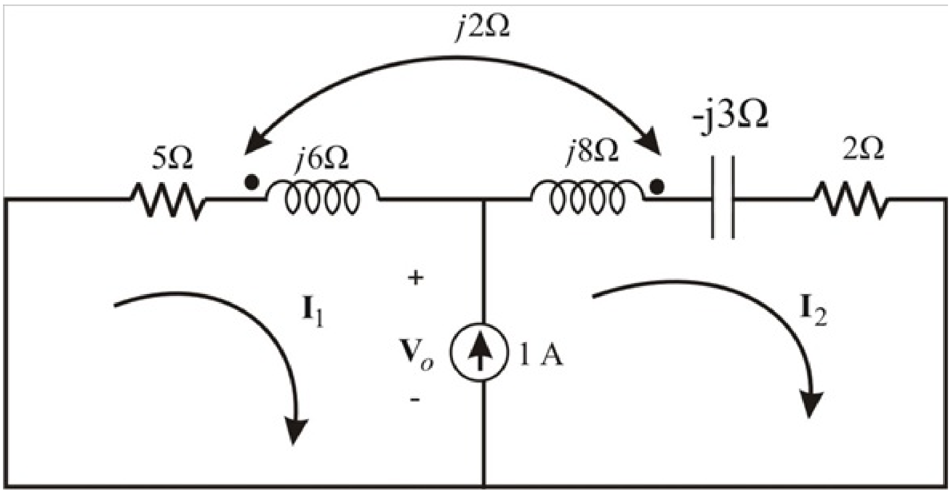


Figure 2

Write the super mesh equation.

$$\mathbf{I}_2 - \mathbf{I}_1 = 0$$
$$\mathbf{I}_2 = \mathbf{I}_1 + 1$$

Apply Kirchhoff's voltage law around outer loop.

$$(5+j6)\mathbf{I}_1 - j2\mathbf{I}_2 + j8\mathbf{I}_2 - j2\mathbf{I}_1 + (2-j3)\mathbf{I}_2 = 0$$
$$\mathbf{I}_1(5+j4) + \mathbf{I}_2(2+j3) = 0$$

Substitute $\mathbf{I}_1 + 1$ for $\mathbf{I}_2$ in the equation.

$$\mathbf{I}_1(5+j4) + (\mathbf{I}_1+1)(2+j3) = 0$$
$$\mathbf{I}_1(5+j4+2+j3) + 2 + j3 = 0$$
$$\mathbf{I}_1(7+j7) = -(2+j3)$$
$$\mathbf{I}_1 = 0.3642\angle -168.69° \text{ A}$$

Comments (2)

Anonymous

Shouldn't the first constraint equation be I_2-I_1 = 1A??

Anonymous

^You're right

Apply Kirchhoff's voltage around left side loop.

$$\mathbf{V}_o + (5+j6)\mathbf{I}_1 - j2\mathbf{I}_2 = 0$$
$$\mathbf{V}_o + (5+j6)\mathbf{I}_1 - j2(\mathbf{I}_1 + 1) = 0$$
$$\mathbf{V}_o + (5+j4)\mathbf{I}_1 - j2 = 0$$
$$\mathbf{V}_o - \frac{(2+j3)(5+j4)}{7+j7} - j2 = 0$$
$$\mathbf{V}_o = 1.5 + j3.786\ \Omega$$

Thevenin's impedance is

$$\mathbf{Z}_{th} = \frac{\mathbf{V}_o}{1\text{ A}}$$
$$= \frac{1.5 + j3.786}{1\text{ A}}$$

Therefore, the Thevenin's impedance $\mathbf{Z}_{th}$ is $\boxed{1.5 + j3.786\ \Omega}$.

Comments (1)

Anonymous

Since, there's no dependent sources, can't you just find Z_th by the sum of the left parallel to the sum of the right??

The Thevenin’s equivalent circuit is shown in Figure 3.

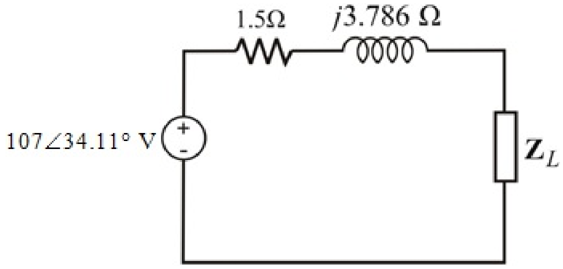


Figure 3

Therefore, the Thevenin’s equivalent circuit is shown in Figure 3.